

# Sanghoon Lee

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Research Group for Bio-Embodied Physical AI

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 <https://huno130.github.io/>

 Google Scholar ·  GitHub ·  LinkedIn

*I unify AI, middleware, and control to realize Embodied Intelligence under uncertainty and resource constraints.*

For full publications, open-source projects, talks, and an interactive research map, see [huno130.github.io](https://huno130.github.io).

## RESEARCH INTERESTS

**Cyber-Physical Artificial Intelligence (CPAI)** — unifying AI, middleware, and control to realize Embodied Intelligence under uncertainty and resource constraints. Particular depth in the **ROS 2 core and its middleware internals** (DDS — FastDDS, CycloneDDS — and Zenoh): QoS semantics, discovery (SPDP/SEDP), the RTPS wire protocol, executors, and observability.

- **AI.** Physical AI, Multi-Robot Systems, Autonomous Mobile Robots, Deep Reinforcement Learning
- **ROS 2.** Core / middleware internals (RTPS, DDS, Zenoh), QoS & Discovery, Observability, Next-gen Middleware, Cross-layer Optimization, Software-Defined Robotics
- **CPS.** Wireless Control Systems, Cyber-Physical Security, Resilient Architectures, Vulnerability Analysis

## EDUCATION

| Period            | Institution                                       | Degree / Program                                                                                                                                |
|-------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022.02 – 2026.02 | DGIST, Daegu, South Korea                         | Ph.D. in Electrical Engineering and Computer Science<br><br>Combined Program (UG-G.). Major GPA 4.3 / 4.3<br><br>Advisor. Prof. Kyung-Joon Park |
| 2019.06 – 2019.09 | University of California, Los Angeles (UCLA), USA | International Student Program. Major GPA 4.3 / 4.3                                                                                              |
| 2018.03 – 2022.02 | DGIST, Daegu, South Korea                         | B.S. in Computer Science. Major GPA 3.99 / 4.3                                                                                                  |

## EXPERIENCE

| Period            | Role                                  | Affiliation                                         |
|-------------------|---------------------------------------|-----------------------------------------------------|
| 2026.02 – Present | InnoCore Postdoctoral Research Fellow | Research Group for Bio-Embodied Physical AI, DGIST  |
| 2022.02 – 2026.02 | Graduate Research Assistant           | Cyber-Physical Systems Integration (CSI) Lab, DGIST |
| 2020.08 – 2022.02 | Undergraduate Research Student        | Cyber-Physical Systems Integration (CSI) Lab, DGIST |

| Period            | Role                     | Affiliation                                           |
|-------------------|--------------------------|-------------------------------------------------------|
| 2020.01 – 2020.02 | Research Intern (Winter) | Computer Architecture and Systems (CAS) Lab,<br>DGIST |

## PH.D. DISSERTATION

### Cyber-Physical Artificial Intelligence. A Unified Framework for Embodied Intelligence

Sanghoon Lee. Advisor. Prof. Kyung-Joon Park.

DGIST, Feb. 2026.

<https://scholar.dgist.ac.kr/handle/20.500.11750/59636>

## INTERNATIONAL PUBLICATIONS

*33 international publications and manuscripts, including 18 first or co-first authored.*

*7+ additional papers currently in preparation.*

*\* indicates equal contribution as first authors.*

### Under Review (2026)

#### [U14] **Cyber-Physical AI. Unifying Cyber-Physical Systems and Physical Artificial Intelligence**

**Sanghoon Lee**, Guoliang Xing, Xue Liu, Karl H. Johansson, Kyung-Joon Park.

*Invited Paper. ACM Transactions on Cyber-Physical Systems (TCPS), 20th Anniversary Special Issue (TCPS-2026-0101). Under Review.*

#### [U13] **Harness Engineering for Physical AI. Robot Middleware Is the Harness Layer**

**Sanghoon Lee**, Jiyeong Chae, Kyung-Joon Park.

ACM/IFIP/USENIX International Middleware Conference (Middleware) 2026 (#256). *Under Review.* [arXiv:2606.09416](https://arxiv.org/abs/2606.09416).

#### [U12] **ros2probe. Non-intrusive, Kernel-selective Observability for Robot Operating System 2 Middleware**

Jisang Yu\*, **Sanghoon Lee\***, Yeonwoo Choi, Kyung-Joon Park.

ACM/IFIP/USENIX International Middleware Conference (Middleware) 2026. Research Paper (#228). *Under Review.* [arXiv:2606.10746](https://arxiv.org/abs/2606.10746) · [Project Page](#).

#### [U11] **The Three Dimensions of ROS 2 Middleware**

**Sanghoon Lee**, Taehun Kim, Angelo Corsaro, Kyung-Joon Park.

ACM Computing Surveys (CSUR). *Under Review.* [arXiv:2607.01304](https://arxiv.org/abs/2607.01304).

#### [U10] **From Hype to Reality. A Survey of Physical AI in Industrial Robotics**

**Sanghoon Lee\***, Jiyeong Chae\*, Jinhong Park, Kyung-Joon Park.

Engineering Applications of Artificial Intelligence (EAAI). *Major Revision.*

#### [U9] **Dependency Chain Analysis of ROS 2 DDS QoS Policies. From Lifecycle Tutorial to Static Verification**

**Sanghoon Lee**, Junha Kang, Kyung-Joon Park.

IEEE Internet of Things Journal (IoT-J). *2nd Minor Revision (IoT-62530-2026.R1).* [arXiv:2509.03381](https://arxiv.org/abs/2509.03381).

#### [U8] **Probabilistic Latency Analysis of the Data Distribution Service in ROS 2**

**Sanghoon Lee**, Hyung-Seok Park, Jiyeong Chae, Kyung-Joon Park.

IEEE Transactions on Vehicular Technology (TVT). *Under Review.* [arXiv:2508.10413](https://arxiv.org/abs/2508.10413).

#### [U7] **Optimizing ROS 2 Communication for Wireless Robotic Systems**

**Sanghoon Lee**, Taehun Kim, Yeonwoo Choi, Kyung-Joon Park.

IEEE Internet of Things Journal (IoT-J). *Under Review.* [arXiv:2508.11366](https://arxiv.org/abs/2508.11366).

#### [U6] **LIMA. Local Intersection Marshaling Algorithm for Multi-Agent Path Finding in Large-Scale Industrial Environments**

**Sanghoon Lee\***, Jisang Yu\*, Geon-pyo Kim, Kyung-Joon Park.  
Robotics and Autonomous Systems (RAS), Elsevier. *Under Review*. [SSRN](#).

- [U5] **Exploiting ROS 2's Blind Spot. Cross-Layer Attacks from Discovery to ARP Spoofing**  
Hyunho Ryu\*, **Sanghoon Lee\***, Young-Sik Kim, Dan Dongseong Kim, Kyung-Joon Park.  
IEEE Internet of Things Journal (IoT-J). *Under Review* (IoT-67776-2026).
- [U4] **Natural-Language to Operational Policies. LLM-Driven Costmaps for Industrial AMRs**  
Jiyeong Chae, Hyunkyo Seo, Heonjae Lee, **Sanghoon Lee**, Kyung-Joon Park.  
Robotics and Autonomous Systems (RAS). *Major Revision*.
- [U3] **Robust Localization in Dynamic Indoor Environments Using Structure-Guided Point-wise LiDAR Scoring**  
Yeong-gi Hong, Donghyung Lee, **Sanghoon Lee**, Jiyeong Chae, Kyung-Joon Park.  
IEEE Sensors Journal. *Under Review*.
- [U2] **Characterizing DDS Discovery Bursts in ROS 2 Multi-Robot Systems**  
Taehyun Kim, **Sanghoon Lee**, Kyung-Joon Park.  
International Conference on ICT Convergence (ICTC) 2026. *Under Review*.
- [U1] **Reliability-Aware 5G-V2X Network Topology Optimization at Signalized Intersection for Edge Learning**  
Seobin Jeong, **Sanghoon Lee**, Kyung-Joon Park.  
IEEE Open Journal of Vehicular Technology (OJ-VT). *Under Review*.

## Published

- [19] **Discovery Storm. Scalability Analysis of DDS and Zenoh in Large-Scale Wireless Robotic Networks**  
**Sanghoon Lee**, Yeonwoo Choi, Jiyeong Chae, Kyung-Joon Park.  
IEEE INFOCOM NetRobICS 2026 Workshop, Feb. 2026.
- [18] **Time-aware Costmap for Smoother and Less Disruptive AMR Navigation With ROS 2**  
Jiyeong Chae, Hyunkyo Seo, **Sanghoon Lee**, Kyung-Joon Park.  
International Journal of Control, Automation, and Systems (IJCAS), Dec. 2025.  
<https://link.springer.com/article/10.1007/s12555-025-0558-8>
- [17] **DIDoS. Disturbance-induced Denial-of-Service Attack in Networked Cyber-Physical Systems**  
Sangjun Kim, **Sanghoon Lee**, Kyung-Joon Park.  
International Journal of Control, Automation, and Systems (IJCAS), Nov. 2025.  
<https://link.springer.com/article/10.1007/s12555-025-0572-x>
- [16] **Vulnerability Analysis of Cross-Layer Attacks on ROS 2**  
Hyunho Ryu, **Sanghoon Lee**, Kyung-Joon Park.  
International Conference on ICT Convergence (ICTC) 2025, Oct. 2025.  
<https://csi.dgist.ac.kr/HOME/csi/data/upload/2025101621472768foe95fa4555.pdf>
- [15] **Poster. How to Send Large Data in ROS 2**  
**Sanghoon Lee**, Taehun Kim, Jiyeong Chae, Kyung-Joon Park.  
IEEE International Conference on Network Protocols (ICNP) 2025, Sep. 2025.  
<https://ieeexplore.ieee.org/abstract/document/11192433/>
- [14] **From Issues to Routes. A Cooperative Costmap with Lifelong Learning for Multi-AMR Navigation**  
Jiyeong Chae\*, **Sanghoon Lee\***, Hyunkyo Seo, Kyung-Joon Park.  
Journal of Industrial Information Integration (JIII), Sep. 2025.  
<https://www.sciencedirect.com/science/article/abs/pii/S2452414X25001645>
- [13] **Integrating ROS 2 and Physical AI. Architecture and Challenges**  
**Sanghoon Lee**, Jiyeong Chae, Kyung-Joon Park.

International Conference on Ubiquitous and Future Networks (ICUFN) 2025, Jul. 2025.  
<https://ieeexplore.ieee.org/abstract/document/11170038/>

- [12] **An Analytical Latency Model of the Data Distribution Service in ROS 2**  
Hyung-Seok Park, **Sanghoon Lee**, Kyung-Joon Park.  
IEEE International Conference on Computer Communications (INFOCOM) 2025, May 2025.  
<https://ieeexplore.ieee.org/abstract/document/11044454/>
- [11] **PINMAP. A Cost-efficient Algorithm for Glass Detection and Mapping Using Low-cost 2D LiDAR**  
Jiyeong Chae, Hyunkyo Seo, **Sanghoon Lee**, Kyung-Joon Park.  
IEEE Transactions on Instrumentation and Measurement (TIM), May 2025.  
<https://ieeexplore.ieee.org/document/10991965>
- [10] **Cyber-Physical AI. Systematic Research Domain for Integrating AI and Cyber-Physical Systems**  
**Sanghoon Lee**, Jiyeong Chae, Kyung-Joon Park.  
ACM Transactions on Cyber-Physical Systems (TCPS), Feb. 2025.  
<https://dl.acm.org/doi/10.1145/3721437>
- [9] **Priority-Driven Resource Allocation with Reuse for Platooning in 5G Vehicular Network**  
Tae-Woo Kim, **Sanghoon Lee**, Dong-Hyung Lee, Kyung-Joon Park.  
Sustainability, Feb. 2025.  
<https://www.mdpi.com/2071-1050/17/4/1747>
- [8] **Learning-enabled Flexible Job-Shop Scheduling for Scalable Smart Manufacturing**  
Sihoon Moon, **Sanghoon Lee**, Kyung-Joon Park.  
Journal of Manufacturing Systems (JMS), Sep. 2024.  
<https://www.sciencedirect.com/science/article/pii/S0278612524002139>
- [7] **False Alarm Prevention through Domain Knowledge-driven Machine Learning. Leakage Detection in Water Distribution Networks**  
**Sanghoon Lee\***, Jiyeong Chae\*, Sihoon Moon, Kyung-Joon Park.  
IEEE Sensors Journal, Aug. 2024.  
<https://ieeexplore.ieee.org/abstract/document/10643479/>
- [6] **Learning-Enabled Network-Control Co-design for Energy-Efficient Industrial Internet of Things**  
Sihoon Moon, **Sanghoon Lee**, Wonhong Jeon, Kyung-Joon Park.  
IEEE Transactions on Network and Service Management (TNSM), Apr. 2024.  
<https://ieeexplore.ieee.org/abstract/document/10285130>
- [5] **Deep Reinforcement Learning-driven Scheduling in Multijob Serial Lines. A Case Study in Automotive Parts Assembly**  
**Sanghoon Lee**, Jinyoung Kim, Yongsoon Eun, Kyung-Joon Park.  
IEEE Transactions on Industrial Informatics (TII), Feb. 2024.  
<https://ieeexplore.ieee.org/document/10210628>
- [4] **A Survey and Perspective on Industrial Cyber-Physical Systems (ICPS). From ICPS to AI-augmented ICPS**  
Jiyeong Chae\*, **Sanghoon Lee\***, Junhyung Jang, Seohyung Hong, Kyung-Joon Park.  
IEEE Transactions on Industrial Cyber-Physical Systems (TICPS), Oct. 2023.  
<https://ieeexplore.ieee.org/abstract/document/10285426>
- [3] **Graph-based Reinforcement Learning for Flexible Job Shop Scheduling with Transportation Constraints**  
Sihoon Moon, **Sanghoon Lee**, Kyung-Joon Park.

Annual Conference of the IEEE Industrial Electronics Society (IECON) 2023, Oct. 2023.  
<https://ieeexplore.ieee.org/document/10312647>

- [2] **Real-Time Controller Reconfiguration for Delay-Resilient Cyber-Physical Systems**  
Sangjun Kim, **Sanghoon Lee**, Kyung-Joon Park.  
IEEE Access, Oct. 2022.  
<https://ieeexplore.ieee.org/abstract/document/9896851>
- [1] **Resilient Controller Reconfiguration under Network Delay Attack in Cyber-Physical Systems**  
Sangjun Kim, **Sanghoon Lee**, Kyung-Joon Park.  
International Conference on Electronics, Information, and Communication (ICEIC) 2022.

## DOMESTIC PUBLICATIONS

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- [D4] **End-to-End Latency Analysis of Reliable Communication in ROS 2 over Wireless Multi-hop Networks**  
Taehyun Kim, **Sanghoon Lee**, Kyung-Joon Park.  
Korean Institute of Communication and Information Sciences (KICS) Summer Conference, Jun. 2025. *Best Paper Award*.
- [D3] **Channel Modeling and SDN-based Routing Path Design for Satellite-Terrestrial Integrated Networks**  
Hyunho Ryu, **Sanghoon Lee**, Kyung-Joon Park.  
Korean Institute of Communication and Information Sciences (KICS) Summer Conference, Jun. 2024.
- [D2] **SDN Based Multipath Routing Algorithm in Satellite-Terrestrial Integrated Network**  
Haewon Jeon, **Sanghoon Lee**, Dong-Hyung Lee, Changhee Joo, Kyung-Joon Park.  
Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference, Apr. 2024.
- [D1] **Sound-based Machine Fault Diagnosis via Active Learning in Industrial Environments with Labeling Constraints**  
Jiyeong Chae, **Sanghoon Lee**, Kyung-Joon Park.  
Korean Institute of Communication and Information Sciences (KICS) Winter Conference, Jan. 2024.

## BOOK IN PREPARATION

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**ROS 2 System Handbook.** A standard reference textbook on ROS 2 systems engineering. *In preparation*.

## PATENTS

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- [P3] **Method and Apparatus for Generating Indoor Maps**  
Jiyeong Chae, Hyunkyo Seo, **Sanghoon Lee**, Kyung-Joon Park.  
Republic of Korea, Application No. 10-2024-0145250, Oct. 22, 2024. *Filed*.
- [P2] **Method and Apparatus for Detecting Leakage in Water Pipeline Networks**  
Young-gi Hong, Kyung-Joon Park, **Sanghoon Lee**, Jiyeong Chae.  
Republic of Korea, Application No. 10-2025-0008593, Jan. 21, 2025.
- [P1] **Leakage Detection Device and Method**  
Kyung-Joon Park, **Sanghoon Lee**, Jiyeong Chae.  
Republic of Korea, Application No. 10-2024-0004960, Jan. 11, 2024.

## HONORS AND AWARDS

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- [9] **Early-Career Researcher Academic Research Grant**, Korean Institute of Communication and Information Sciences (KICS). 2026.
- [8] **Best Paper Award**, 2025 KICS Summer Conference. 2025.

- [7] **Next-Generation Engineering Researcher**, Institute for Promotion of Engineering and Science of Korea (IPESK). 2025.
- [6] **Excellence Award in Future Tech Division**, Aero Future Tech & Business Idea Contest, Hanwha Aerospace. *Team Leader*. 2024.
- [5] **Selected as 2023 Top 100 Excellent National R&D Achievements**, Ministry of Science and ICT (MSIT), Republic of Korea. *Participant Researcher*. 2023.
- [4] **ACM MobiHoc Student Grant**, ACM SIGMOBILE. 2022.
- [1–3] **Dean's List (Academic Excellence Award)**, DGIST. Fall 2021, Spring 2021, Spring 2020.

## INVITED TALKS AND CONFERENCE PRESENTATIONS

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| Date            | Venue                                 | Type                                                                                      |
|-----------------|---------------------------------------|-------------------------------------------------------------------------------------------|
| 2026.10.14 – 16 | ICTC 2026, Sheraton Jeju, South Korea | <b>Invited Talk</b> — "ROS 2. A Robot Middleware for Physical AI" (tentative)             |
| 2026.09.22 – 24 | ROSCon 2026, Toronto, Canada          | <b>20-min Talk</b> — "ROS 2 Middleware Survival Guide. Navigating Space, Time, and State" |
| 2026.07.20      | Medisby HQ, Seoul, South Korea        | Invited Seminar — Physical AI for Rehabilitation Robotics                                 |
| 2026.03.23      | DOGU                                  | Invited Seminar — ROS 2 Architecture Optimization for Multi-Robot Systems                 |
| 2026.03.09      | KETI                                  | Invited Seminar — Multi-Robot + Fleet Management Middleware (DDS, QoS, observability)     |
| 2026.01.28      | Holiday Robotics                      | Invited Seminar                                                                           |
| 2026.01.21      | ROSCon KR 2026, Seongnam              | Session Speaker                                                                           |

## SERVICE TO ROBOTICS COMMUNITY

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**ROSCon Korea 2027 Organizing Committee Member.** Co-host institutions PinkLAB · XYZ · DGIST. Co-chair Prof. Kyung-Joon Park (DGIST). Held at DGIST, Feb. 17–19, 2027. Site <https://roscon.ros.org/kr/2027/>

## PROFESSIONAL SERVICE — PEER REVIEW

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### 2026 – Present

- *Springer Nature*. Artificial Intelligence Review. Computing
- *IEEE*. Transactions on Systems, Man, and Cybernetics

### 2025 – Present

- *Springer Nature*. Scientific Reports. International Journal of Computational Intelligence Systems
- *IEEE*. Journal of Communications and Networks (JCN). Systems Journal. Transactions on Network and Service Management (TNSM)
- *Elsevier*. Expert Systems With Applications. Computers and Operations Research

### 2024 – Present

- *Springer*. Machine Learning
- *Elsevier*. ICT Express
- *IEEE*. Transactions on Industrial Cyber-Physical Systems (TICPS)

### 2023 – Present

- *IEEE*. Access

## OPEN-SOURCE SOFTWARE

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### **QoS Guard** — ROS 2 DDS QoS Static Validator

Offline verification of 16 QoS policies across Discovery / Data Exchange / Disassociation phases with detection of 40 common constraint violations. Companion to publication [U9].

<https://github.com/csi-dgist/QoS-Guard> · [Project Page](#) · [arXiv:2509.03381](#)

### **DDS Optimizer for Wireless Large Payload Transfer** — ROS 2 Wireless Communication Optimizer

XML-based QoS configuration framework for ROS 2 DDS over wireless. Caps RTPS message size at 1472 B, optimizes heartbeat rates, and sizes HistoryCache to prevent buffer bursts. Companion to publication [U7].

<https://github.com/csi-dgist/DDS-Optimizer-for-Wireless-Large-Payload-Transfer> · [arXiv:2508.11366](#)

### **ros2probe** — Host-Level Observability for ROS 2 DDS Traffic

eBPF-based passive monitoring tool that captures RTPS/DDS packets and reconstructs the full ROS graph, topic metrics, and message streams in userspace, with zero subscriptions and no middleware intrusion. SEDP event-driven data-centric GID synchronization (C3) and adaptive visibility (C4) provide non-trivial novelty. Rust implementation. CLI `rp` / GUI / MCAP recorder / Unix-socket JSON-RPC. Validated on Laptop, NVIDIA Jetson Orin series, Raspberry Pi 4/5. Companion to publication [U12].

<https://github.com/csi-dgist/ros2probe> · [Project Page](#)

## TECHNICAL SKILLS

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**Systems** — C++, Python, Rust, Linux (Kernel / Driver Analysis), eBPF, Docker, Git

**Middleware** — ROS 2, DDS (FastDDS, CycloneDDS), Zenoh, TCP/IP, UDP

**Intelligence** — PyTorch, TensorFlow, Nav2, MoveIt 2, Cartographer, AMCL

**Simulation** — NVIDIA Isaac Sim, MuJoCo, Gazebo, MATLAB, NS-3, OMNeT++

## CITATIONS AND INDICES

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Google Scholar. 250+ citations (Jul. 2026). h-index 9. i10-index 9.

*Last updated. July 2026.*